

Lecture 2 Introduction to Linux Notes

1. What is an Operating System?

An **operating system** provides software features of a computer that are fundamental. It will allow to use the computer hardware providing the basic tools that make the device useful.

2. What is a kernel?

A **Kernel** is responsible for managing low-level features of the computer. For example, memory allocation, CPU time, managing the system hardware, and program to program interaction.

3. Which other parts aside from the kernel identify an OS?

Graphical User Interfaces identify an OS because it allows user to interact with the device through icons, menus, and mouse pointer. **Command Line shells** was used before graphical user interfaces which is used by typing commands. Tools like **web browsers**, **document processors** and **text editors**.

4. What is Linux?

Linux is an operating system that is Unix-like Operating System. It has libraries, utilities, and a kernel that make up its entire operating system. **Linux** is also available is a lot of distributions, for example, Arch, Red Hat, Slackware, Ubuntu, and etc.

5. What is a Linux distribution?

Linux distribution system is a complete operating system that is tailored around the **Linux** kernel. Other elements are Core Unix Tools, Supplemental software, startup scripts, and installer.

6. List at least 4 Linux characteristics:

1. *Linux* is a open source software that is available to everyone
2. Free of charge
3. *Linux* is scalable and customizable
4. Majority of servery applications run on versions of *Linux*

7. What is Debian?

Debian is an organization that is all-volunteer, which dedicate to develop and promote free software. It is to be known as the grandfather of all **Linux** distributions.

8. List and define the different types of licensing agreements

1. Open source - can be distributed for free or charge. Able to be modified or add ons.
2. Closed source - software not distributed with the source code, not allowed to be modified.
3. Free software - software is distributed with the source code

9. What is Free Software? Define the 4 freedoms.

0. Freedom0 : use the software for any purpose
1. Freedom1 : examine the source code and modify it as seen fit
2. Freedom2 : redistribute the software
3. Freedom3 : redistribute your modified software

10. What is virtualization?

Defined as creating a virtual version of something. Used to let multiple operating systems run on one physical machine at the same time, it also allows admins to divide the hardware to be able to create multiple computers into a singular physical computer.